

# EQUIPMENT SUBMITTAL

## Clean Agent System

**Project:** Meridian Data Centre Campus - Phase 1

**Reference Section:** 21 22 00

**Package ID:** SUB-212200-01-R0

**Revision:** R0

**Reviewed Spec Revision:** Rev 2024

**Date:** 01-01-2025

ACTIVE

| Rev | Date       | Description                 |
|-----|------------|-----------------------------|
| R0  | 01-01-2025 | Initial / Updated Submittal |

Technical Parameters

| Parameter                        | Claimed Value  |
|----------------------------------|----------------|
| Suppressant Agent Type           | FK-5-1-12      |
| Total Clean Agent Agent Quantity | 11800 kg       |
| Cylinder Operating Pressure      | 360 psi        |
| Design Concentration             | 4.7%           |
| NOAEL Safety Limit               | 10%            |
| Standards Reference              | NFPA 2001-2008 |

<sup>1</sup> discharge time extends to 11.0 seconds when piping runs exceed 45 meters  
<sup>2</sup> Discharge Time is 9.2 s (standard pipe configuration)

| Clause ID             | Requirement Summary                                                                                     | Response                 |
|-----------------------|---------------------------------------------------------------------------------------------------------|--------------------------|
| 21 22 00 Part 1.3.3   | Factory test and compliance certificates issued by UL and FM for all clean agent storage cylinder va... | Comply                   |
| 21 22 00 Part 1.2.1   | The clean agent fire suppression system shall be designed, installed, and tested in accordance with ... | NFPA 2001 (2008 Edition) |
| 21 22 00 Part 2.2.3.A | Clean agent discharge time shall not exceed 10 seconds from the moment the discharge valve opens unt... | Comply                   |
| 21 22 00 Part 2.2.3.B | Agent hold time (retention) shall be a minimum of 10 minutes (600 seconds) to prevent re-ignition. H... | 9 minutes                |
| 21 22 00 Part 2.3.4.F | The vendor shall submit a NOAEL safety margin calculation demonstrating that the actual maximum agen... | Comply                   |
| 21 22 00 Part 2.3.1.A | Clean agent storage cylinders shall be pressurised to 500 psi (34.5 bar) with nitrogen super-pressur... | Comply                   |
| 21 22 00 Part 2.2.5   | Minimum total clean agent quantity for the campus shall be 12,500 kg of FK-5-1-12. Agent quantity pe... | Comply                   |
| 21 22 00 Part 1.2.3   | All clean agent storage cylinders and discharge valve assemblies shall be tested, listed, and label...  | ISO 9809-1               |
| 21 22 00 Part 2.1.1   | The clean agent shall be FK-5-1-12 (Dodecafluoro-2-methylpentan-3-one), chemically identified as CF3... | Comply                   |
| 21 22 00 Part 2.2.1   | Design concentration shall be 4.7% by volume (v/v) for Class A and Class C fire hazards. The design ... | Comply                   |

| Clause ID           | Requirement Summary                                                                                     | Response |
|---------------------|---------------------------------------------------------------------------------------------------------|----------|
| 21 22 00 Part 2.2.2 | The No Observed Adverse Effect Level (NOAEL) for FK-5-1-12 is 10.0% by volume (v/v). The design conc... | Comply   |